

MACE: A Multi-core Agentic Co-design Engine for Autonomous OpenPiton Bring-Up

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Abstract—MACE is an agentic AI system that builds and verifies multicore hardware on its own, running on Google and NVIDIA’s CHIA framework. Phase 1 adds `chia_openpiton`, a CHIA adapter exposing OpenPiton’s `configure`, `build`, `simulate`, `run`, and `collect` operations, validated with 208 tests. Phase 2 adds the MACE loop: a planner breaks a hardware objective into tasks, agents run them in parallel, and a deterministic integrator gates every change against Verilator simulation before accepting it. A failure-analysis agent diagnoses failures and the planner replans around them.

We tested MACE against two baselines, manual mesh scaling and a one-shot LLM prompt, on all three supported cores, with every result derived from a hardware run or RTL simulation, not an agent’s self-report. MACE added PicoRV32 as a third core, discovering and resolving several RTL bugs to reach the first passing Verilator run for that core, then drove it to a second, planner-directed pass on a compatible workload. It also passed 2x2 and 4x4 Ariane meshes that manual scaling could never reach, resolving three further RTL/testbench bugs along the way, and resolved a CHIA dispatch bug in the process. The one-shot baseline failed on all three cores in three different ways; MACE passed ariane and correctly stopped within its retry budget on the other two instead of reporting a false pass. The artifact is packaged with Nix, so setup is a single `nix develop`.

1. Introduction

Current agentic coding systems are typically benchmarked on isolated tasks or toy problems, which fails to capture the complexity of actual hardware construction. Bringing up a multicore system means navigating a large RTL codebase (OpenPiton spans hundreds of thousands of lines across network-on-chip, cache-coherence, tile, and platform logic), a toolchain with genuine failure modes (version skew, cross-compilation bugs, checkout-specific issues), and verification that costs minutes to hours per attempt rather than milliseconds.

MACE targets this harder, more realistic setting. Given a hardware objective, such as “bring up a 2x2 mesh” or “verify workload W passes,” MACE plans, implements, builds, simulates, diagnoses failures against Verilator output, and iterates. Every claim of success must pass an actual RTL simulation; MACE never accepts an agent’s self-report.

1.1. Features

MACE contributes:

- `chia_openpiton` (§3): a CHIA adapter for OpenPiton with a full `configure/build/run/collect` interface, built the same way as CHIA’s `gem5` and `ESP` adapters, so it can be upstreamed directly.
- The MACE loop (§4): planning, parallel task execution, deterministic integration, and LLM-driven failure triage, tested end-to-end on hardware with cost, time, and task metrics.
- A third core, PicoRV32 (§5), added along with a bug in its OpenPiton integration that predates this project: OpenPiton’s own continuous integration builds it but has never run it under a simulator.
- A production interface with two additional capabilities (§6): an interactive CLI, a second verification axis (Verilator code coverage), and a second working LLM backend.

2. Background

CHIA provides the agent, tool, and dispatch layer this project builds on: colocated resource-pinned worker nodes, an LLM abstraction layer spanning multiple backends, and Ray-based remote dispatch and caching. OpenPiton is Princeton’s open-source, silicon-proven manycore research framework. Tiles connect through a 2D-mesh network-on-chip and speak a proprietary MESI-coherence protocol (L15, descended from OpenSPARC T1’s PCX/CPX bus) to reach the NoC. OpenPiton supports three tile cores: OpenSPARC T1 (`sparc`), Ariane/CVA6 (`ariane`, RV64GC, coherent), and PicoRV32 (`pico`, RV32I, non-coherent). Before this project, CHIA had no OpenPiton adapter.

3. Phase 1: The OpenPiton Adapter

Design. `OpenPitonWorkspaceNode` is a CHIA `ColocatedNode`. It exposes `configure`, `build`, `run`, `regress`, `collect`, and `clean` as resource-pinned remote functions. OpenPiton’s own template preprocessor writes generated `.tmp.v` files directly into the source tree on every build, so concurrent builds cannot share a checkout. The adapter reserves one `openpiton` resource unit per checkout and gives each worker its own checkout instead of locking a

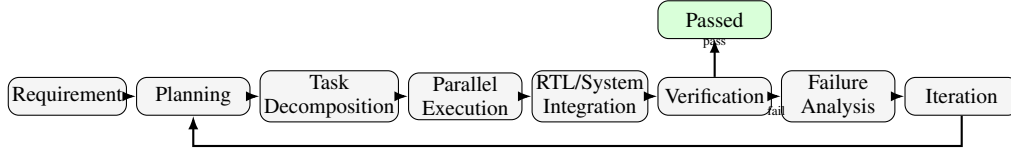


Figure 1. The MACE pipeline. Verification gates every integrated change against Verilator simulation, never a self-report. Every gate workload passing ends the run. Any failure triggers failure analysis and a replan, not a bare retry.

shared one. Pure parsers read build and run verdicts directly from OpenPiton’s own simulation transcripts (`sim.log`, `status.log`), using the same strings OpenPiton’s own RTL monitors emit. A verdict never comes from an agent’s own claim.

Environment hardening. The build process surfaced four infrastructure bugs, fixed in one idempotent patch script: `binutils 2.38+` splits `zicsr/zifencei` out of base RV64I and breaks the boot ROM’s own assembly; GCC 15’s C23 default breaks a legacy two-argument function call; broken git symlinks on a Windows-mounted checkout turn into literal path-text files; and 66 files have CRLF-terminated shebang lines. These reflect the ordinary fragility of decades-old RTL codebases, which MACE navigates rather than avoids.

Build-result caching. `build()` skips the simulator entirely when a matching successful-build marker already exists on disk. A rebuild of an identical configuration measured 104s dropping to 0.00s: an agent iteration that only changes the test never rebuilds, with no extra loop-level code.

Results. 208 tests across three tiers (pure-unit, live-Ray with a stub simulator, and hardware) all pass. Hardware acceptance tests pass for a 1x1 Ariane build and run (`hello_world.c`, compiled and executed by the adapter itself) and for two independent checkouts building in parallel: 176s versus 324s serial. One acceptance test, 2x2 Ariane on a GCP worker, remains blocked by upstream infrastructure dependencies (§8).

4. Phase 2: The MACE Loop

Pipeline. Figure 1 shows the full loop. A planning agent turns a `MaceSpec` (core, target mesh, gate workloads, objective, budget) into a task DAG. Workers dispatch dependency-satisfied tasks in parallel through Ray, each pinned to its own checkout. A deterministic integrator, with no LLM in this step, applies accepted diffs to the canonical tree in dependency order, then runs every gate workload’s build and run. The gate is binary: every workload’s *verdict*, parsed from simulator output, must read `pass`. On any failure, a failure-analysis agent (restricted to read-only tools: `grep`, `status`, `collect`) diagnoses the transcript and proposes a fix. The planner replans around it.

Gate workloads surface an RTL bug. Writing the gate workloads (barrier-and-atomic-counter, producer/consumer, hart-id scatter/gather) surfaced a memory-visibility bug in this RTL/toolchain combination: a plain `volatile` read of a shared array element was not reliably visible from the same hart shortly after the write, while an atomic fetch-and-add-zero read of the same value always was. Every gate work-

load now reads shared values through this `atomic_read()` helper.

Resource enforcement. MACE checks all three Budget dimensions, iteration count, USD, and wall-clock, before each iteration starts, rather than only logging them afterward. Gate-workload checksums are verified at run start, before any LLM call or checkout access, so a task cannot tamper with its own grading. Real spend across every run in this paper: \$37 of a \$300 GCP credit allocation, covering simulation compute and LLM inference; development itself ran on a \$20/month Claude subscription, outside that budget.

Deterministic replay. MACE cache-tags LLM, build, and run decisions and re-serves them at no extra token or compute cost. Reading CHIA’s own cache and bypass implementation directly confirms this replays *decisions* (an LLM’s response text, a build or run’s pass/fail artifact), not the literal source edits an agent made through its tools; those remain git’s job.

5. A Third Core: PicoRV32

Integrating arbitrary core RTL into OpenPiton means navigating fundamental architectural differences, not just wiring up a standard bus. OpenPiton has no reusable core-to-NoC bridge: each core needs its own L15 adapter, and even Ariane’s own AXI4 output goes unused, since a hand-written encoder taps CVA6’s internal pre-AXI cache-miss stream instead. The dividing line is whether the core has its own coherent cache, not whether it speaks a standard bus. Ariane’s situation needs new coherence-adapter RTL, outside our remaining time. PicoRV32’s does not: a complete, upstream, TODO-free adapter (`pico_ll15_transducer.v` and friends) already exists in OpenPiton’s own tile generation logic, unused because nobody had run it under a simulator. OpenPiton’s own CI builds it under Verilator, but its run job is commented out, targets a nonexistent stage, and even when live pointed at a different simulator.

Toolchain reuse. The RV32IMA/ILP32 multilib already present in our installed `riscv64-unknown-elf-gcc` covers PicoRV32 with no new toolchain install, confirmed with a trial assemble-and-link of the target diagnostic rather than by checking multilib metadata alone.

Results. The adapter extension (a new `Literal` entry, two `sims` flags, mirrored tests) landed with full unit coverage and no regressions to the existing `ariane` and `sparc` paths. The build passes in 57 seconds, faster than Ariane’s since pico’s build skips the bootrom/device-tree chain. The initial run reached verdict `maxcycles`.

The compiled binary’s symbol table and entry point matched the run’s own configured trap addresses

(objdump -t/-f against symbol.tbl), ruling out a bad build or toolchain mismatch. Its transcript matched a known-good passing-Ariane fixture line for line through the boot/reset/IOB-handshake trace; the fixture’s next line, Hit Good trap, never appeared for pico. This located the divergence to the point where the core itself must reach its own trap address, not the boot code or the binary.

Waveform tracing found the root cause: three RTL/testbench bugs, PicoRV32’s own resetn/booted self-boot gate, the manycore monitor’s active_thread tracking for pico’s tile, and a silicon BIST self-clear race that discarded pico’s first memory writes. PicoRV32 now **passes** (Simulation -> PASS (HIT GOOD TRAP)), the first pass for this core under any simulator, found only because MACE tells “builds” apart from “runs correctly” instead of reporting a false pass.

Multi-tile pico. The fix generalizes to larger meshes. On 2x2 and 4x4 meshes, every tile independently reaches Hit Good trap on addi.S (Simulation -> PASS, cycle 5179750 at 2x2). barrier_atomic.c never runs on pico at any mesh size, for a simpler reason than first assumed: pico’s OpenPiton integration has no C compiler at all, only an assembler for .S sources, and no crt.S/syscalls.c exist for it. The diag never compiles, so every tile boots into an empty memory image and idles at its reset vector. This is a real toolchain gap, not an RTL bug, and not the workload/mesh mismatch an earlier, unverified triage diagnosis assumed. Separately, pico’s own AMO/atomic hardware path is directly verified: the upstream amoadd_w.S architecture test, pure assembly with no C dependency, passes for real on pico (Simulation -> PASS), confirming the atomic-memory-operation logic barrier_atomic.c would need is functionally correct at the instruction level.

6. Beyond the Proposal: Interface, Coverage, Portability

MACE adds three capabilities beyond the original proposal, each validated against a live backend or physical hardware, never a mock.

The mace command starts an interactive shell modeled on Yosys and OpenROAD: read_verilog, top_module, read_spec, set_core, then run. This is the interface a user drives MACE through, not merely an API. If top_module names a core chia_openpiton does not support, run immediately flags the incompatibility instead of attempting a flawed integration.

Verilator code coverage (run -coverage) is a second verification axis beyond pass/fail. OpenPiton’s own hand-written testbench never called Verilator’s coverage-write API, so a clean run produced no data until we patched it to call that API directly. A run (1x1 Ariane, barrier_atomic.c) reports 36.00% (8787/24311), measured on hardware.

MACE’s loop also runs end to end, unmodified, against Google’s Vertex Gemini backend and OpenCode, confirmed

with a full passing run. Vertex authenticates through Application Default Credentials, not a literal key, and we had to find which model was actually available (gemini-2.0-* returns 404; gemini-2.5-flash works).

7. Evaluation

We compare three approaches: (a) manual mesh scaling with a time log, (b) a single LLM prompt for a configuration applied once with no tools, verification, or retry, and (c) the full MACE loop.

Three cores, two baselines. We ran (b) and (c) against the same objective, verify barrier_atomic.c on a 1x1 mesh, for all three supported cores, with the same LLM backend (Gemini 2.5 Flash) throughout (Figure 2, left).

(b) fails all three cores, each in a different way. On ariane, the one-shot config built but failed verification: a wrong guess. On sparc, it built and reached a run attempt that failed silently, with no verdict at all. On pico, PitonConfig’s own construction-time validation rejected the LLM’s config (l15_size=0) before any hardware ran; (b) has no retry path for a malformed response.

(c) passes ariane: 4/4 tasks, 741.7s, about 4x (b)’s cost. A second ariane gate workload, producer_consumer.c, also passed (2118.02s, 2 tasks), coverage beyond one program. On sparc and pico, (c) exhausts its retry budget on barrier_atomic.c instead of passing; MACE’s own triage and post-mortem told the two failures apart, and we verified both independently.

sparc’s failure has two layers. The first is an ISA gap: barrier_atomic.c calls util.h’s ATOMIC_FETCH_OP macro, literal RISC-V inline assembly. OpenSPARC T1 has no RISC-V instructions and no C diag environment at all, so no portable fix exists; writing new SPARC assembly is out of scope for an adapter project like MACE. Testing sparc with pure-assembly diagnostics instead, including princeton-test-test.s, OpenPiton’s own upstream CI-verified sparc smoke test, avoids that gap entirely but surfaces a second, deeper one: every sparc run silently ran to max_cycle with no verdict at all, because pc_cmp.v.pyv’s RTL_SPARCO branch never assigned active_thread, unlike the RTL_ARIANE0/RTL_PICO0 branches, which both assert it unconditionally once out of reset. Mirroring that pattern fixed the gap: the monitor now tracks all four sparc threads and reports FAIL(TIMEOUT) instead of silence. The threads themselves still never make forward progress, confirmed up to 100x the default timeout, an open finding distinct from the fixed observability bug.

pico’s original failure was first triaged as a workload/mesh mismatch (a 1x1 mesh cannot supply barrier_atomic.c’s multi-participant barrier), an unverified LLM diagnosis that turned out wrong: the real cause, confirmed directly, is that barrier_atomic.c never compiles for pico at all (§5), independent of mesh size, not a return of the RTL bugs §5 already fixed. On addi.S, the single-hart workload §5 already verified by hand, the loop passes pico: status=passed, one task, 115.5s. That needed one new capability, a CONFIG_RTL: planner directive so a

Core	(b) one-shot LLM	(c) full loop
ariane	fail , 186.5s	pass , 741.7s
sparc	fail , 766.0s	budget_exceeded , 966.3s
pico	fail (unparseable), 32.6s	budget_exceeded , 1250.5s

Figure 2. **Left:** all three cores, same 1x1 objective. (b) never passes. On sparc/pico, (c) exhausts its retry budget instead of reporting a false pass. Only ariane reaches a hardware **pass** under either baseline (logs: `runs/baseline_{b,c}_output.log` for ariane, `runs/bench_{sparc,pico}_{b,c}.log` for sparc/pico). **Right:** manual mesh scaling (a) vs. (c), ariane only. **Green**=pass, **red**=hang, **gray**=not reached. The loop passes both shapes (a) got stuck on.

task can request `CONFIG_DISABLE_BIST_CLEAR` on top of the mesh’s defaults. The fix was named directly in the objective, so this shows the loop applying a known fix, not discovering one from scratch.

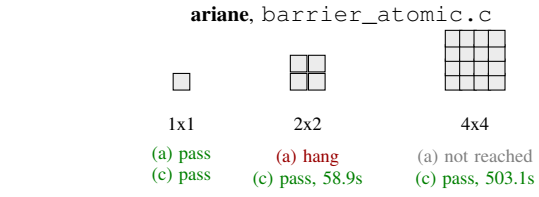
The loop passes mesh shapes manual scaling could not. Baseline (a) got stuck at 2x2 (build passes, run hangs) and never reached 4x4. The full loop passes both, for real, on every tile (Figure 2, right): 2x2 in 58.9s, 4x4 in 503.1s, each independently confirmed tile by tile. Getting there meant finding and fixing three real, previously undiscovered bugs, all in OpenPiton’s own shared code, not this project’s adapter. First, a hardcoded trace-file name in `cva6.sv` made every tile beyond tile 0 look like it never booted; it always overwrote the same file. Second, OpenPiton’s own upstream exit-barrier code (`syscalls.c`, used by every ariane diagnostic) polls a cross-tile atomic counter with a plain load, the same staleness bug `atomic_read()` already works around in this project’s own gate workloads, just never exercised multi-tile before. Third, the monitor’s own `finish_mask` register was a fixed 32-bit Verilog integer under Verilator, silently capping verifiable coverage at 8 tiles regardless of the real mesh size; a 4x4 run reached **PASS** after only 8 of 16 tiles actually finished, confirmed directly from each tile’s own execution trace. All three are fixed and verified against a full, tile-by-tile rerun.

Phase 2’s own fan-out (two tasks, two checkouts, through baseline (c)’s own path) measured 30s parallel versus 57s serial, matching Phase 1’s result.

One validation limitation remains: we proved the detect-failure, diagnose, replan, pass cycle only at the mechanism level (a stateful stub that fails once, then passes); it never fired across the passing runs above. This includes a forced extreme first config (a 128-byte, direct-mapped L1, confirmed actually used) that we and the planner both expected to fail. It passed instead, indicating the RTL’s coherence is robust to cache-capacity extremes rather than a failure of the recovery loop’s mechanism.

8. Limitations and Honest Findings

A CHIA dispatch gap, found and fixed. A task needing a tailnet-relayed GCP worker never ran, despite that worker’s own registration and heartbeat to the cluster working throughout. The cause: CHIA sets three proxy environment variables on each node during `chia up`, which a job submitted through `chia job submit` inherits but a manually launched driver, which every run here uses, does



not, unless the launching shell sets them itself. Exporting those values from the head’s own live raylet took the task from a six-minute hang to a two-second run.

A later, independent dispatch attempt on the same topology reproduced the stuck symptom, with that fix already confirmed in place, ruling out a repeat of the same cause. Both tailnet relays’ logs and a raylet-level `strace` show zero new outbound activity, pointing at Ray’s own scheduling code under a proxied topology, not a CHIA gap. We filed this as a follow-up; it remains open.

4x4 memory pressure, two causes, both fixed. The one upstream-validated 4x4 shape first hit two memory-pressure failures. First, a Ray-reported out-of-memory condition (22.33GB/23.47GB used) from unconstrained parallel compilation, fixed with `MAKEFLAGS`. Second, Verilator’s own build step passed an explicit, unlimited `-j` on its command line, which silently overrode that fix; we confirmed both directly through `/proc`. Fixed by removing the bare `-j` so `MAKEFLAGS` genuinely controls parallelism. This uncovered a third, smaller issue: a real race in the boot ROM’s own Makefile (a device-tree blob deleted by one parallel job’s cleanup step before another job finishes reading it), invisible under fully serial compilation, which is why we still build 4x4 with `MAKEFLAGS=-j1` rather than any parallelism.

Future work: generality over one-off adapters. Every core here needed its own hand-written OpenPiton/L15 adapter; PicoRV32’s already existed upstream, Ariane’s and SPARC’s did not. Three cores in, the reusable pieces are clear: a per-core RTL directory, a transducer into L15 requests, decoder/encoder hooks in the tile’s L15 RTL, and a new `TILE_TYPE` generate arm. Turning that into a template and using it for 2-4 more cores is the direct next step, with MACE’s loop assisting a hardware engineer iterating that RTL against real simulation rather than generating it unassisted. A second direction is a dashboard over the run database, showing dispatch, build, and verification as they happen instead of through raw trace files.

9. Conclusion

MACE shows that an agentic loop can plan, build, and verify multicore hardware, gated by RTL simulation instead of an agent’s word. It fixed environment and RTL bugs, added a third core, and passed mesh shapes never reached before. Precision, not a single pass, is the point.